

Supplementary Information

Quality assessment of a country-wide bicycle node network with loop census analysis

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This is the supplementary information for the manuscript containing supplementary notes and figures.

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Supplementary Note 1: Data acquisition and preprocessing

Gradient data

Gradient data is taken from the publicly available Digital Elevation Model (DEM) of Denmark [5] and applied to $\Delta = 100$ m segments of all links, where a segment’s gradient is the slope associated with the sea level difference between its start and end point. We define a link’s maximum gradient as the maximum of all gradients of its segments. This segmentation procedure ensures that a link’s maximum gradient is not distorted by its length. We chose a reasonable intermediate Δ as a trade-off: Too large Δ risks maximum gradients getting lost due to averaging, too small Δ risks high maximum gradients captured but over inconsequential lengths.

The choice of Δ is also context-dependent and should be calibrated in future work with empirical observations or feedback from cyclists. For example, for any specific Δ , even if there is an unacceptably large average gradient over one or a few Δ -length segments, some cyclists might be tolerant to dismounting and pushing their bicycles over such stretches.

POI data

We use publicly available POI data [6] which was compiled from several open-source data sets (primarily OpenStreetMap and GeoFA) by the company Septima as a service to Dansk Kyst- og Nature Turisme (DKNT) within the framework of the Danish BNN project. For simplicity, here we restrict ourselves to the three top categories of facilities, services, and attractions. We use a buffer of 250 m to match POIs to links, where each POI can be matched to multiple links. With the matched POIs we define a *POI Diversity* $D_i \in \{0, 1, 2, 3\}$ for each link i as its number of unique POI categories. Further, we make the simplifying assumption that each POI provides a water source, which we call the *Water provision* W_i of a link i defined as $W_i = \min(1, D_i)$. In total, out of 54,129 POIs (facility: 30,119, attraction: 13,881, service: 10,129) the process matched 47,571 POIs (facility: 25,283, service: 11,191, attraction: 11,097).

Network data

Our starting point is a topologically slightly simplified version of the raw BNN data, fetched from GeoFA [4] on October 11, 2024, constituting 25 disconnected network components of Denmark’s 6 main islands Jutland, Zealand, Funen, Lolland-Falster, Bornholm, Longland, and of 19 smaller islands. The initial topological simplification is a continuity-preserving operation with a development version from October 2024 of the *neatnet* Python package [3], which was a necessary precondition for the subsequent analysis as it dropped topological artefacts like parallel lines – see Fig. S11 for an example detail.

In a preprocessing step, we then dropped the 19 smaller islands and ran for each of the remaining 6 main islands the network simplifications of dropping self-loops and dangling nodes (blind ends). We report the dropped parts in Fig. S12 in red. This simplification procedure was necessary to transform the network into a shape that makes a loop-centered analysis possible: First, the BNN data from the 19 dropped small network components constitute only 0.69% of total network length and are so small that a loop-centered analysis would have been trivial as there are (almost) no loops possible. Also the analysis of link lengths and face loops would not have been meaningful due to the limited underlying topography. Second, the simplifications were necessary to drop dangling nodes and self-loops which are not relevant for our loop-centered analysis and could have caused computational issues. This preprocessing step reduced the number of connected components from 25 to 6, the number of nodes and links from 8,487 and 12,216 to 7,170 and 10,814, respectively, and the total length from 29,401 km to 28,215 km. Although the number of links decreased by 11.5%, the total length dropped was only 4.0%.

Supplementary Note 2: Justifying the loop capping at 30 nodes

For the computational reason of exponential combinatorial explosion [1], we limit all loops to at most 30 nodes. In this supplementary note we discuss what information this capping loses, and why it is justified for our analysis.

First and most importantly, we check whether for the “Family with small children” scenario we lose any information. The concrete question is: Are there loops shorter than 20 km that we miss by not searching for loops with more than 30 nodes? The answer is a clear “no”, as reported in Fig. S13: In the range below 20 km, the distribution shifts right with increasing number of nodes and eventually disappears above 25 nodes. Therefore, in the relevant range below 20 km, no loops are overlooked, and our assessment for the “Family with small children” scenario is exact.

Second, we check whether for the other scenarios, “Adult leisure cyclist” and “Family with e-bikes”, we lose any information. Here we show that we indeed lose information, see Fig. S14 as the distribution of loops

shifts right with increasing number of nodes but does not completely disappear even at 30 nodes. In fact, there might be in total a few million loops with over 30 nodes but below 40 km not captured due to the capping. The question is whether the neglected loops are important.

We believe this neglect is justified, for two reasons. First, the average link length is 2.61 km, the median is 2.23 km. A loop with over 30 nodes but shorter than 40 km would need to consist of 30 links below the 0.29-quantile of 1.34 km, on average, which is improbable. Also from a practical planning perspective, it is unlikely that cyclists would want to select such a large number of nodes for their round trips. Second, for the “Adult leisure cyclist” scenario, the number of round trips per node is dominated by the number of links in its neighborhood that have a gradient above 6%. Therefore, most neglected loops have to be in areas with both few such high gradient links and high node densities. However, these are at the same time areas where there is already a high saturation of loops at 8+ loop bits, for example Lolland-Falster. As 8+ is our maximum loop bit class, adding more loops to an 8+ loop bit node would not change anything for our analysis. We therefore believe that the neglect of such many-node loops would not change the results substantially, at least not qualitatively. Of course, this claim should be checked in future work, to whatever extent made possible by computational or algorithmic advancements.

Supplementary Note 3: Computational limitations and approximation of water provision

Our implementation is limited by the use of networkX’s algorithm `simple_cycles()` for generating all simple cycles, which is relatively slow and cannot natively take into account spatial properties from our enriched data such as POIs as it is not developed for data-enriched, spatial networks. Further, it does not allow for exceptions for closed walks and circuits in addition to simple cycles, to be able to account for blind ends and other potential exemptions for repeating nodes or links. Its only parameter is `length_bound`, which allows for a node-length cutoff, but any additional considerations had to be manually coded, for simplicity in Python, which is slow. Although python-igraph implemented its `simple_cycles()` routine in summer 2025, which should be faster than the networkX algorithm, the same limitations persist there. A custom extension of the core algorithms in a low level language like C or Rust would be necessary for the needed speed improvement, which would have been a too involved software engineering task outside the scope of the paper.

Because of above limitations, we were not able to calculate the specific loop property of water provision exactly but needed to resort to an approximation here: Our implementation assumes that whenever a link has a water source, then this water source is located on the center of the link. The inaccuracy introduced by this approximation cancels out on average, but it could introduce some false negatives (water source calculated after slightly above 10 km when in reality below) with the same magnitude of false positives (water source calculated after slightly below 10 km when in reality above). In the worst case, a point of water provision could be shifted half of the length of a link. For median link lengths of 2.23 km the introduced worst possible inaccuracy is 2.23 km (half the link length on both ends of a waterless segment). Nevertheless, assumptions about water provisions are already approximate: every POI is assumed to provide a water source, POIs are matched at a distance of 250 m, and the 10 km maximum distance is an arbitrary normative specification. Therefore, we believe that adding a bit more leeway to the calculation of the 10 km water provision requirement does not change results substantially. A possibility to account for these assumptions conservatively could be to calculate with a reduced maximum distance, changing the 10 km to a lower value.

Supplementary Note 4: Topological properties

The bicycle node network (BNN) of DKNT is a manually designed network. In a wayfinding context, BNNs work best when their nodes are simple forks that necessitate decisions only between going left or right (and no other direction). In short, nodes should have degree 3. Further, the choice of link length limits the possible lengths of face loops and vice versa. Since the stated design principles specify optimal lengths for both, here we check if these specifications are consistent with each other. Next, we measure the empirical topological properties of the designed network, to see how it fits the specifications. Last, here we also aim to understand which regular network could be the best model for a BNN: If there were no geographic constraints, what would be the “platonic ideal” of a BNN? Can we infer implicit design principles from studying the empirical network?

We find that the empirical network’s degree frequencies are: 0 nodes of degree 0 or 1, 139 nodes of degree 2, 6775 nodes of degree 3, 255 nodes of degree 4, 1 node of degree 5. These frequencies yield a degree distribution of $P(2) = 1.9\%$, $P(3) = 94.5\%$, $P(4) = 3.6\%$, $P(5) = 0.01\%$. Therefore, the vast majority of nodes has indeed degree 3, which is consistent with our intuition that nodes should be as simple as possible forks for cyclists, see Fig [SI5a](#). Studying the distribution of nodes per face loop, we also find it centered around 6, albeit with higher variation than the degree distribution, Fig. [SI5b](#).

Given that each node should have degree 3, and that many face loops have around 6 nodes, the most natural model for a BNN is a hexagonal network, see Fig. SI5c. Plugging in the targeted optimal link lengths of 1-5 km yields optimal face loop lengths of 6-30 km, which is a slightly larger range than the specified target of 8-20 km [2], but consistent. It is possible to achieve both specifications exactly though, for example with a staggered square grid, see Fig. SI5d. It is isomorphic to the hexagonal network, however, link lengths are not uniform here but distributed such that $\frac{1}{3}$ of all links (the horizontal links in Fig. SI5d) are twice as long as the rest (the vertical links in Fig. SI5d). Varying link lengths from 1-5 km in the staggered grid leads to face loop lengths of 8-20 km, as specified.

Supplementary Note 5: How well does node density explain loop bits?

Given the circumstantial visual evidence from the loop census analysis that more loops sometimes coincide with more nodes per area, we want to check more rigorously whether or to which extent node density is a good proxy for loop bits. Intuitively it makes sense that more nodes per area can imply more links per area, which would imply more loops. Also, if node density were a good proxy for loop bits, then the costly loop census analysis would not be necessary to perform, and node density could serve as a “good enough”, cheap estimate for determining round trip options.

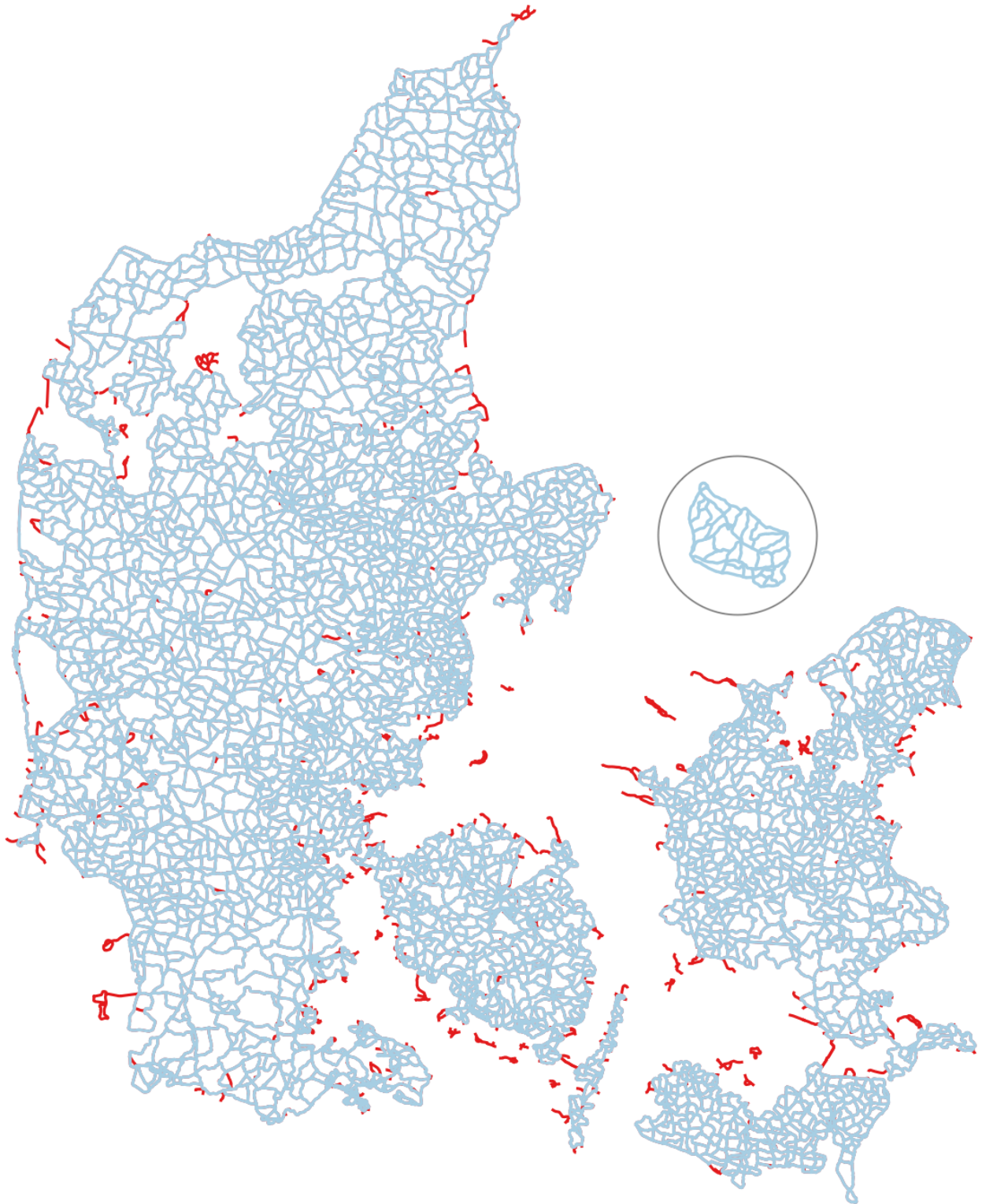
Prior to checking this proxy hypothesis, we assess potential correlations with topographical attributes, performing linear regressions of node density versus geometric link properties (maximum gradient, water provision, POI diversity). To be able to perform regressions, we spatially aggregate these properties via averages on a hexagonal H3 grid with resolution 6, yielding 1,524 grid elements with approximately 3.72 km side length and 36.13 km² area. Figure SI6a–d reports the regressions between node density and average property values over all grid elements. The only relationship we do not correlate is node density versus link length, as this is a clearly non-linear negative association, Fig. SI6a. Given the topological properties of the BNN and its planarity, this association is intuitive, as more nodes imply that there is less space for links. For the regression on the maximum gradient we find the smallest and only non-significant association, at $r = 0.04$ and $p = 0.16$, implying that node density and maximum gradient are not associated linearly. Water provision and POI diversity are significantly associated negatively, with $r = -0.16$ and $r = -0.19$, respectively, and $p < 10^{-9}$.

Finally, to check the hypothesis of node density being a proxy for loop bits, we perform the same regressions as before but for node density versus loop bits over all the three scenarios and their cumulative restrictions. We find all these associations to be positive and significant (all $p < 10^{-9}$), but with varying strengths. As one could expect, the least restricted cases are most strongly associated, Fig. SI6e,i,m, at $r = 0.69$ and $r = 0.74$, respectively. This association could be interpreted as evidence that node density is a good proxy for loop bits, thus for round trips. However, once the maximum gradient restriction is added in the first two scenarios, Figs SI6f,j, the association breaks down to $r = 0.24$ and $r = 0.16$, respectively. The distribution of gradients thus destroys many correlations, weakening the power of node density as a proxy. Further restrictions naturally push the loop bits slightly towards zero, Figs SI6g,h,k,l, and the correlations increase negligibly, to $r = 0.27$, $r = 0.27$, and $r = 0.18$, $r = 0.17$, respectively. Examining the third scenario, Fig. SI6m–p, reveals that most of the correlation collapse comes indeed from the spatial distribution of the maximum gradients, because the water and POI diversity restrictions do not substantially lower the correlations, from $r = 0.69$ only down to $r = 0.68$. To summarize, without a maximum gradient restriction, node density can be a reasonably good proxy for loop bits, but with a maximum gradient restriction the correlation mostly breaks down, at least for Denmark.

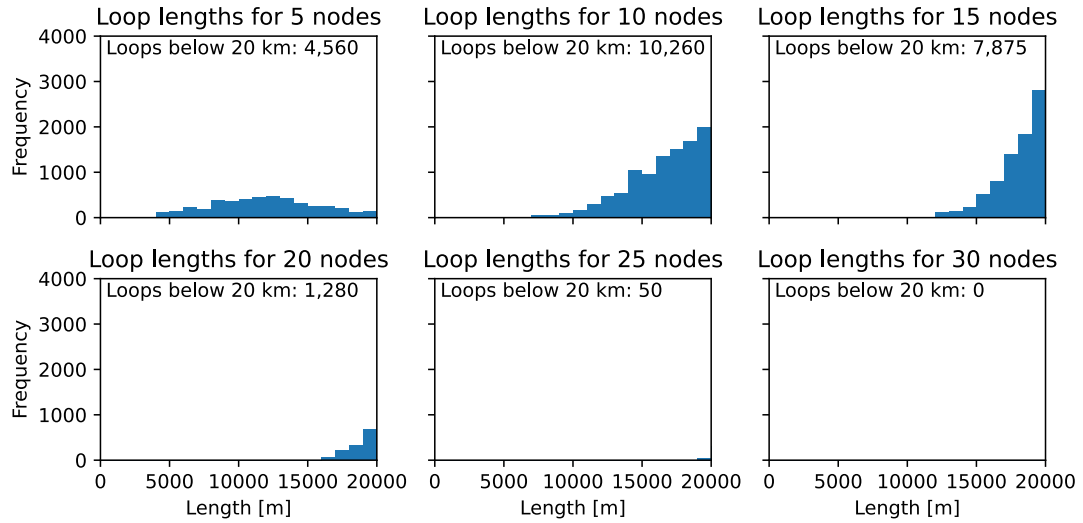
Supplementary Figures



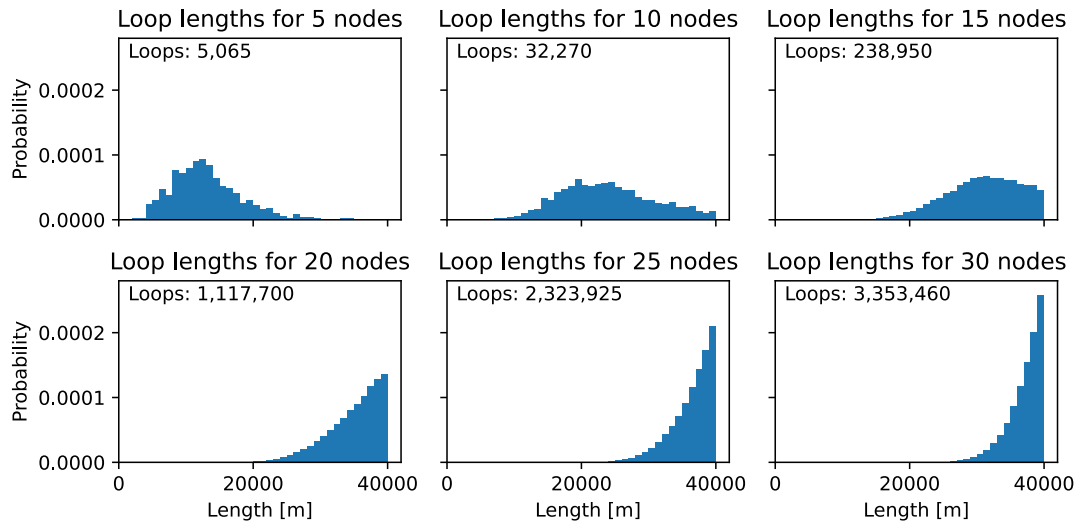
Supplementary Figure S11: **Network detail demonstrating the initial transformation of dropping topological artefacts.** The resulting transformed network is in black. The raw data artefacts (parallel lines) are shown in red.



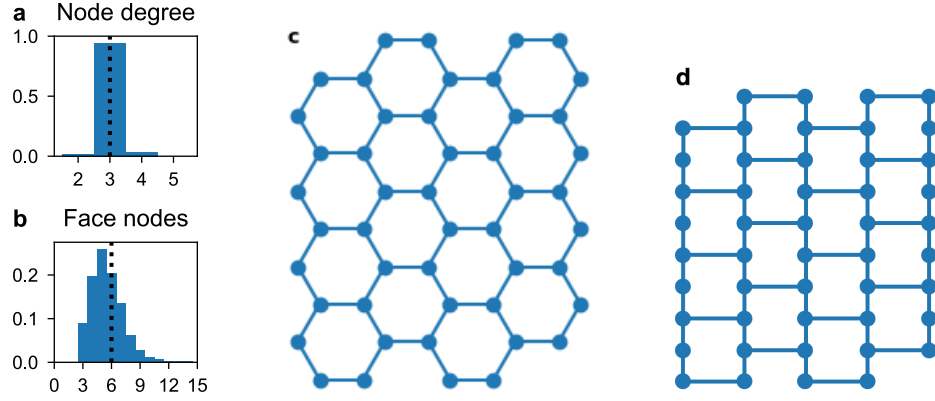
Supplementary Figure SI2: **Network simplification in the preprocessing step.** The dropped parts are shown in red; they consist of small disconnected components, self-loops, and blind ends.



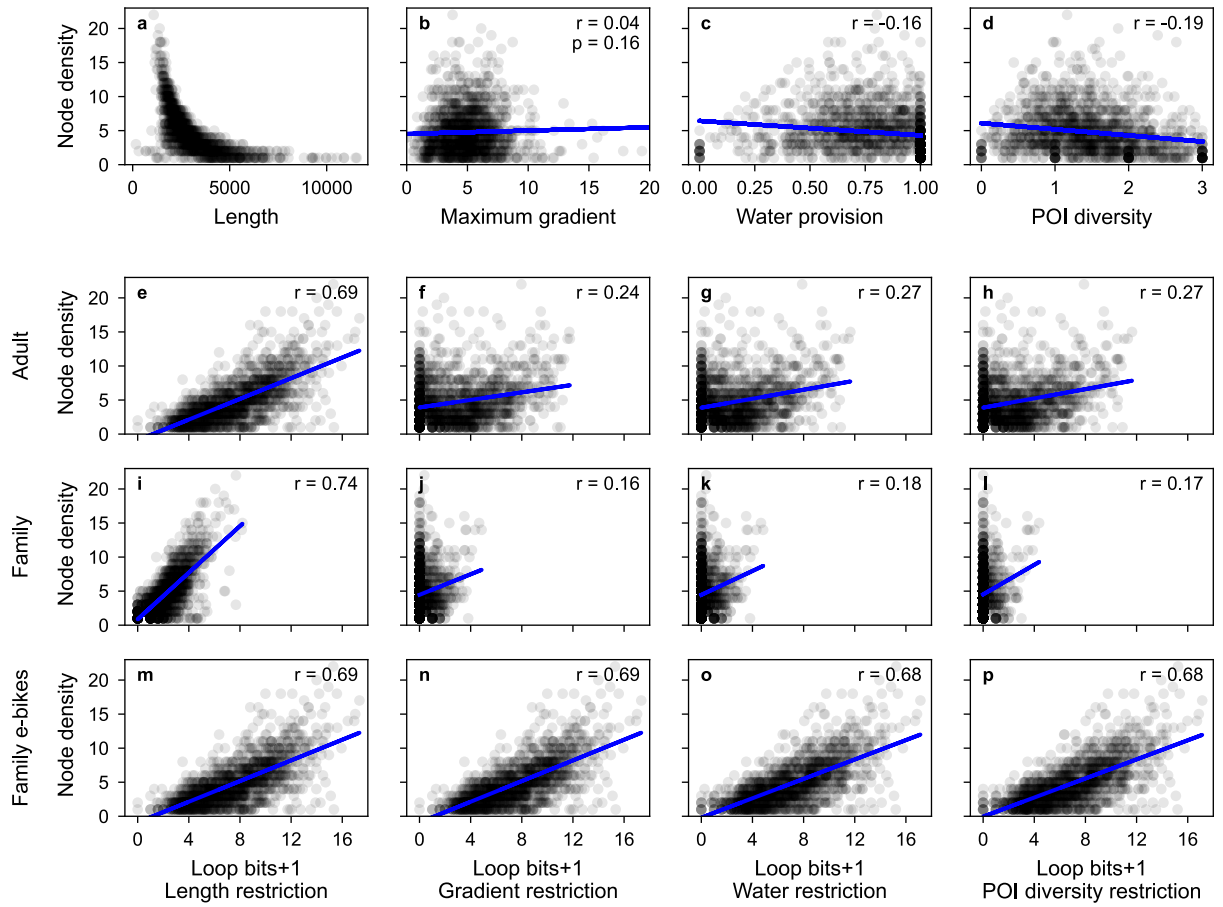
Supplementary Figure SI3: **Frequencies of loop lengths for different numbers of nodes.** In the range below 20 km, the distribution shifts right with increasing number of nodes and eventually disappears above 25 nodes. Therefore, in the range below 20 km, no loops are overlooked, and the assessment for the “Family with small children” scenario is exact.



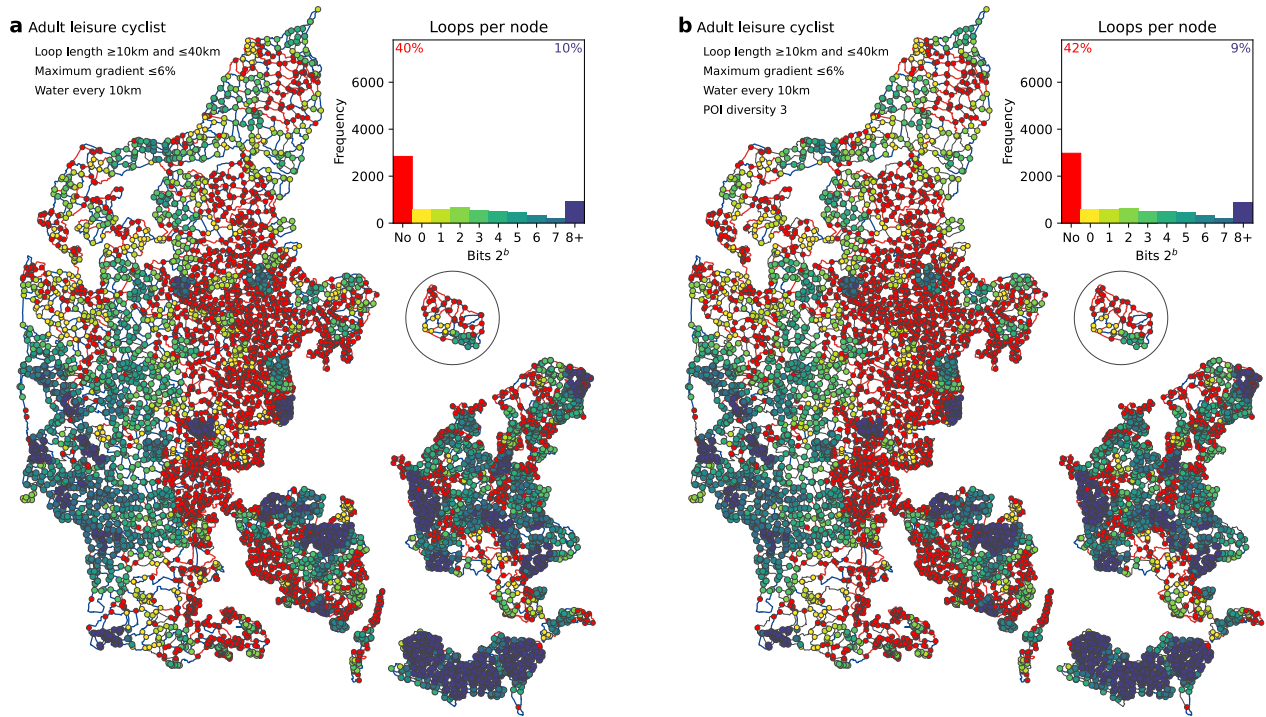
Supplementary Figure SI4: **Probabilities of loop lengths for different numbers of nodes.** In the range below 40 km, the distribution shifts right with increasing number of nodes but does not disappear completely. It is likely that a few million loops are not accounted for in total, especially on the right end of the distribution close to 40 km. Therefore, the assessments for the “Adult leisure cyclist” and “Family with e-bikes” scenarios are underestimating the true number of loops.



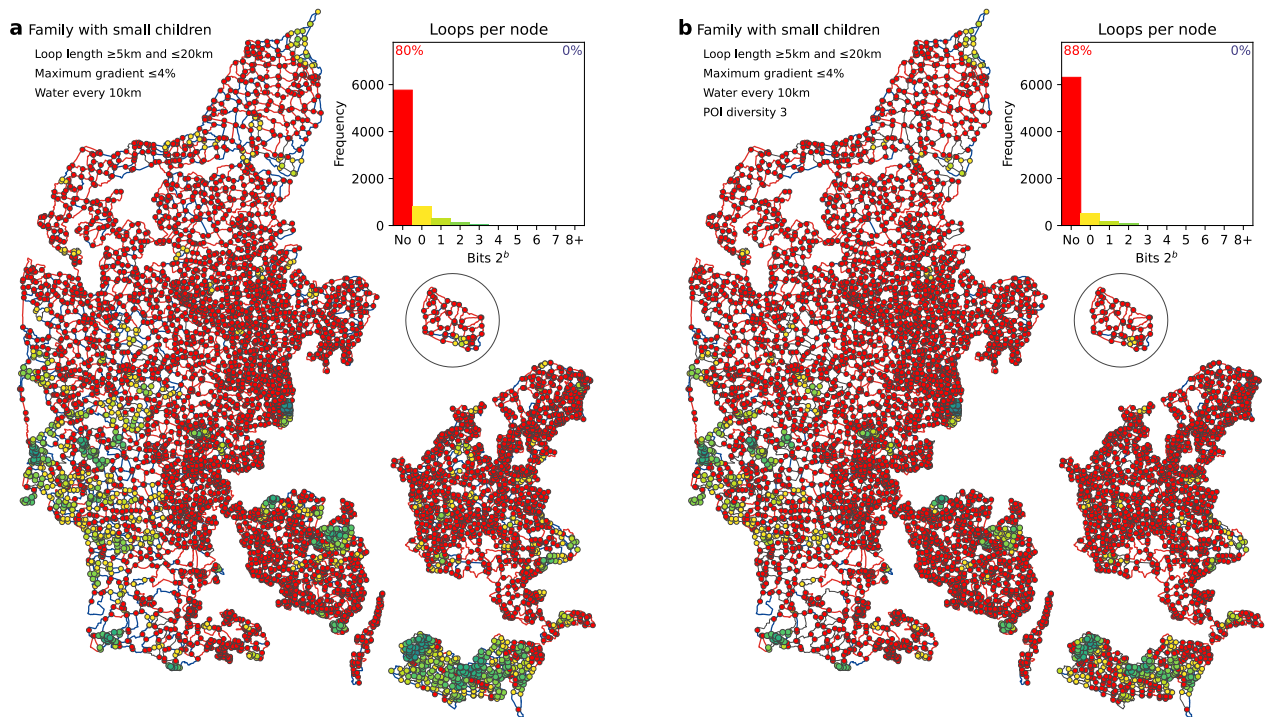
Supplementary Figure SI5: **Topological properties and models for the BNN.** **a.** Degree distribution: The degree of most nodes is 3 (dotted line), implying the prevalence of simple forks where cyclists can choose only between going left or right (and no other direction). **b.** Distribution of nodes per face: The empirical network does not have a constant number of nodes per face, but the distribution is centered around 6 (dotted line). **c.** Given the degree, link length, and face loop length specifications, the network could be modeled as a hexagonal graph. **d.** Alternatively, the network could be modeled as a staggered square grid, isomorphic to the hexagonal graph.



Supplementary Figure SI6: **Node density versus properties.**



Supplementary Figure SI7: **Loops per node for the scenario “Adult leisure cyclist”**. Adding restrictions on top of length and maximum gradient reduces the loops further. **a**. Also restricted by water available every 10 km (blue links), **b**. also restricted by POI diversity 3 (blue links).



Supplementary Figure SI8: **Loops per node for the scenario “Family with small children”**. Adding restrictions on top of length and maximum gradient reduces the loops further. **a**. Also restricted by water available every 10 km (blue links), **b**. also restricted by POI diversity 3 (blue links).

References

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